

CS471 Web Technologies - Laboratory 1 Report

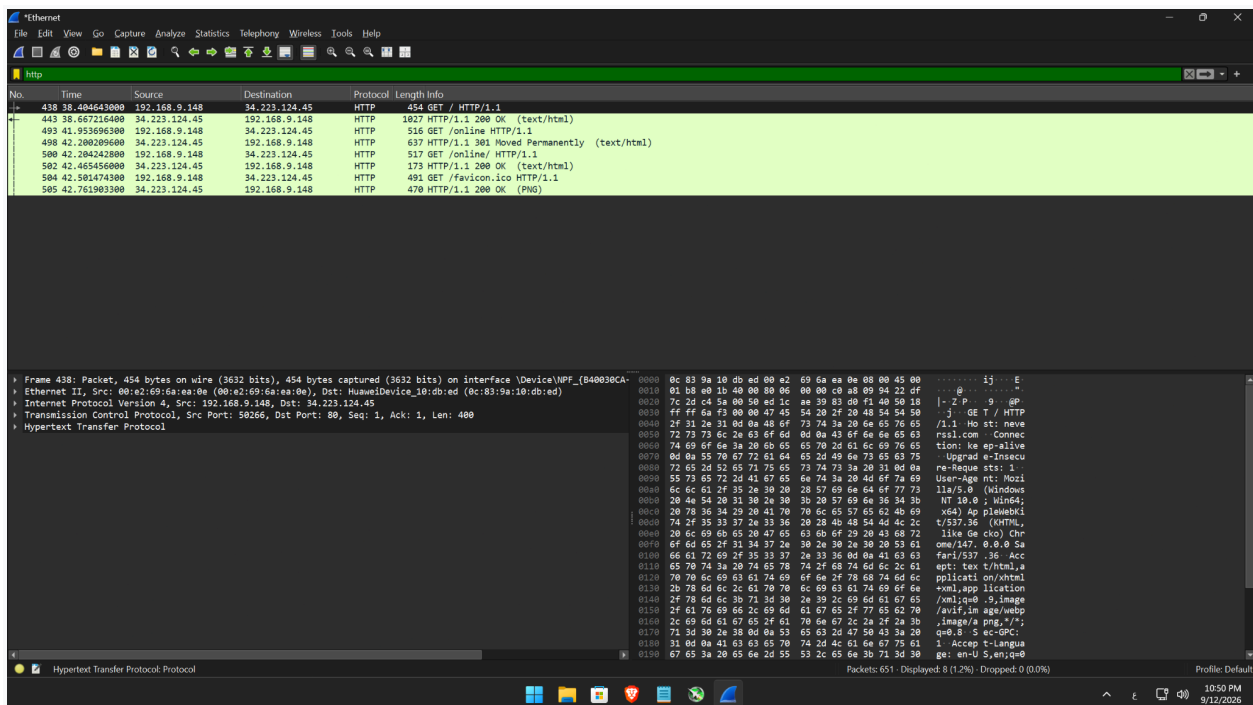
Topic: The Internet Protocols (HTTP, TCP/IP, and UDP Analysis)

Prepared for: College of Computer, Qassim University

Part 1: Capturing and Analyzing HTTP Traffic

In this activity, Wireshark was configured to capture incoming and outgoing packets while accessing an unencrypted web resource (<http://neverssl.com/>). The capture was filtered using the display filter http.

- **HTTP Request:** The client sends an HTTP GET / request (Packet 438) and subsequent GET /online HTTP/1.1 (Packet 493) to the host server.
- **HTTP Responses:** The web server initially responded with HTTP Status Code 301 Moved Permanently, redirecting the client to /online/, followed by HTTP Status Code 200 OK indicating successful content delivery.

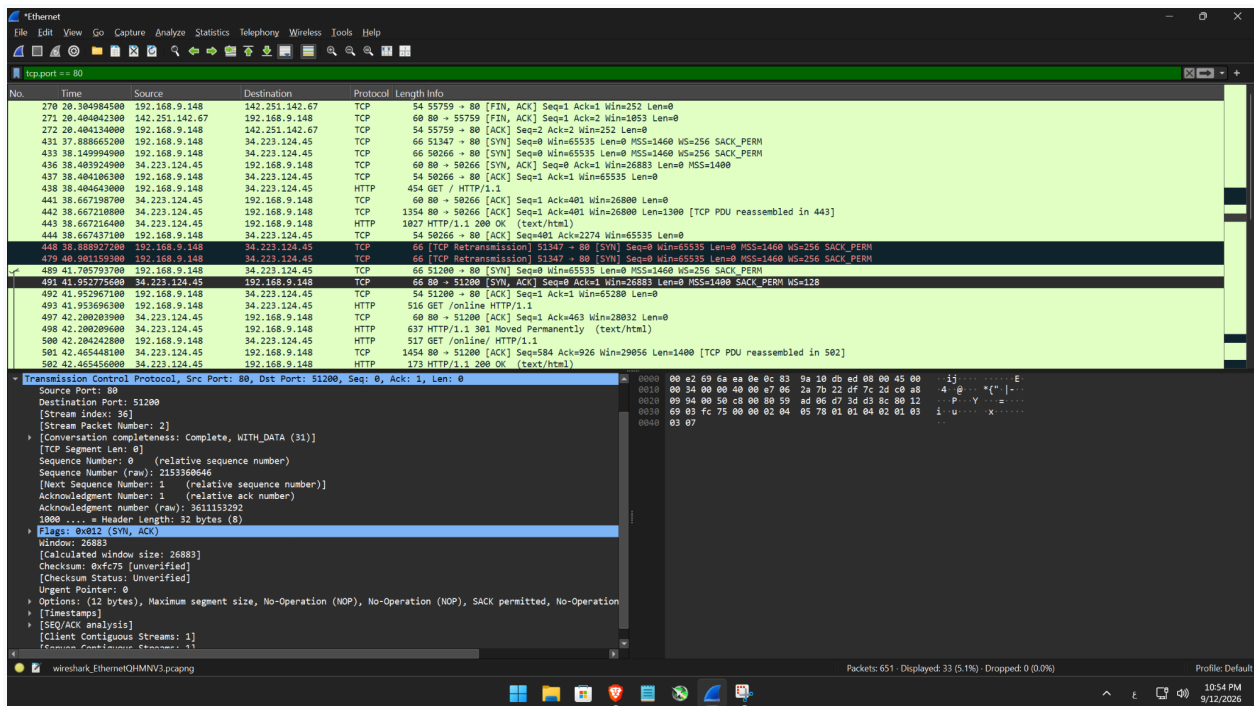


Part 2: Analyzing TCP/IP Traffic

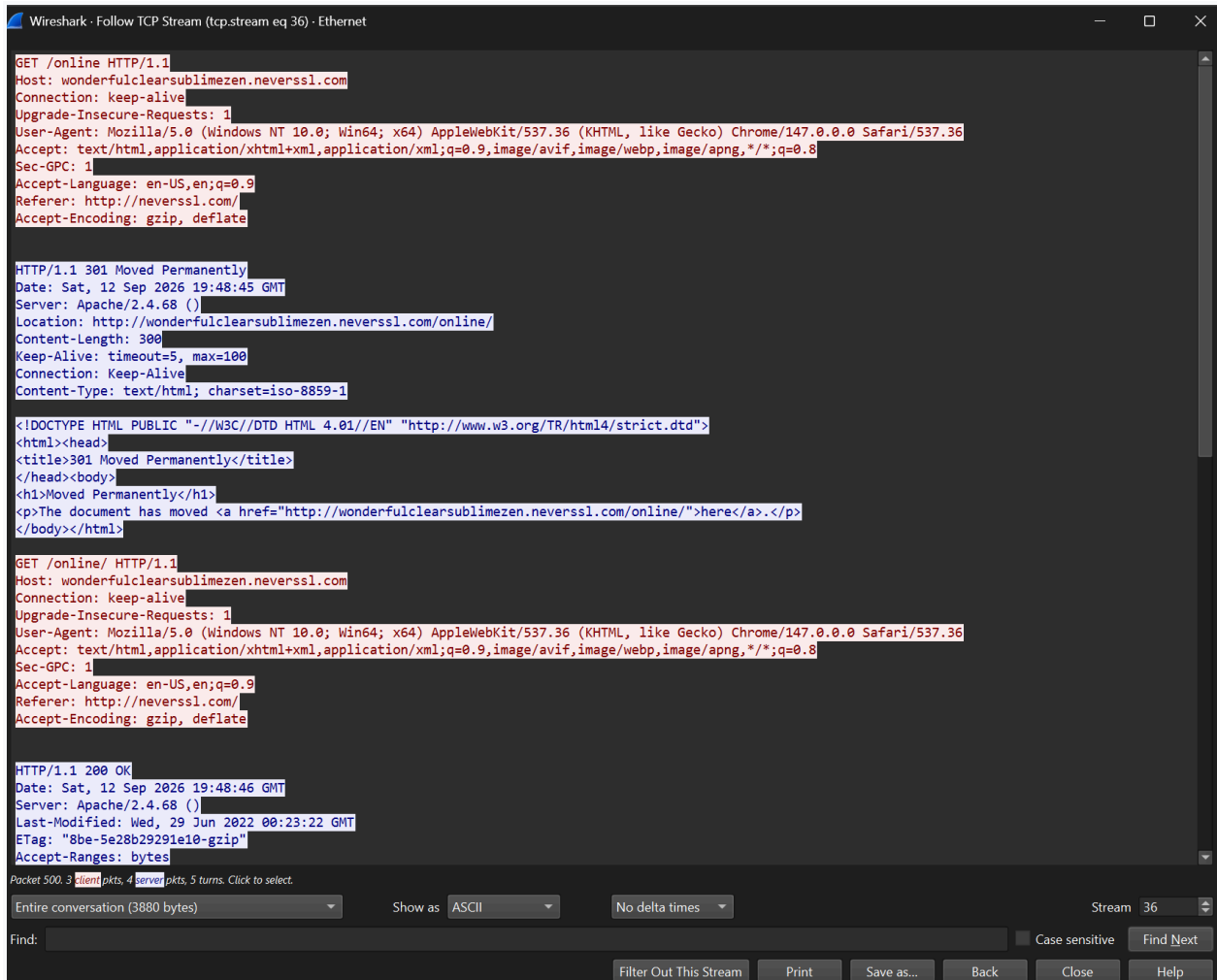
Task 1 & Task 2: Three-Way Handshake and Stream Analysis

TCP is a connection-oriented protocol that establishes a reliable communication session using a Three-Way Handshake before transmitting data:

1. **SYN (Packet 489):** The client initiates the connection by sending a SYN packet with Sequence Number = 0 (relative).
2. **SYN-ACK (Packet 491):** The server responds acknowledging the request with Sequence Number = 0 and Acknowledgment Number = 1 (confirming receipt of client's SYN).
3. **ACK (Packet 492):** The client acknowledges the SYN-ACK by transmitting an ACK packet with Sequence Number = 1 and Acknowledgment Number = 1. The connection is established.



TCP Stream Inspection (Data Transfer): Following the TCP stream revealed the entire client-server dialogue, showing client HTTP request headers (Host, User-Agent, Accept-Encoding) in red and server response headers and HTML payloads in blue.



The screenshot shows the Wireshark interface with the 'Follow TCP Stream' view selected for the Ethernet interface. The selected stream is TCP stream 36. The packet list shows three packets: a GET request (300 bytes), a 301 Moved Permanently response (3880 bytes), and a 200 OK response (3880 bytes). The packet details pane shows the following information:

```
GET /online HTTP/1.1
Host: wonderfulclearsublimezen.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Sec-GPC: 1
Accept-Language: en-US,en;q=0.9
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate

HTTP/1.1 301 Moved Permanently
Date: Sat, 12 Sep 2026 19:48:45 GMT
Server: Apache/2.4.68 ()
Location: http://wonderfulclearsublimezen.neverssl.com/online/
Content-Length: 300
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/html; charset=iso-8859-1

<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html><head>
<title>301 Moved Permanently</title>
</head><body>
<h1>Moved Permanently</h1>
<p>The document has moved <a href="http://wonderfulclearsublimezen.neverssl.com/online/">here</a>.</p>
</body></html>

GET /online/ HTTP/1.1
Host: wonderfulclearsublimezen.neverssl.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Sec-GPC: 1
Accept-Language: en-US,en;q=0.9
Referer: http://neverssl.com/
Accept-Encoding: gzip, deflate

HTTP/1.1 200 OK
Date: Sat, 12 Sep 2026 19:48:46 GMT
Server: Apache/2.4.68 ()
Last-Modified: Wed, 29 Jun 2022 00:23:22 GMT
ETag: "8be-5e28b29291e10-gzip"
Accept-Ranges: bytes
```

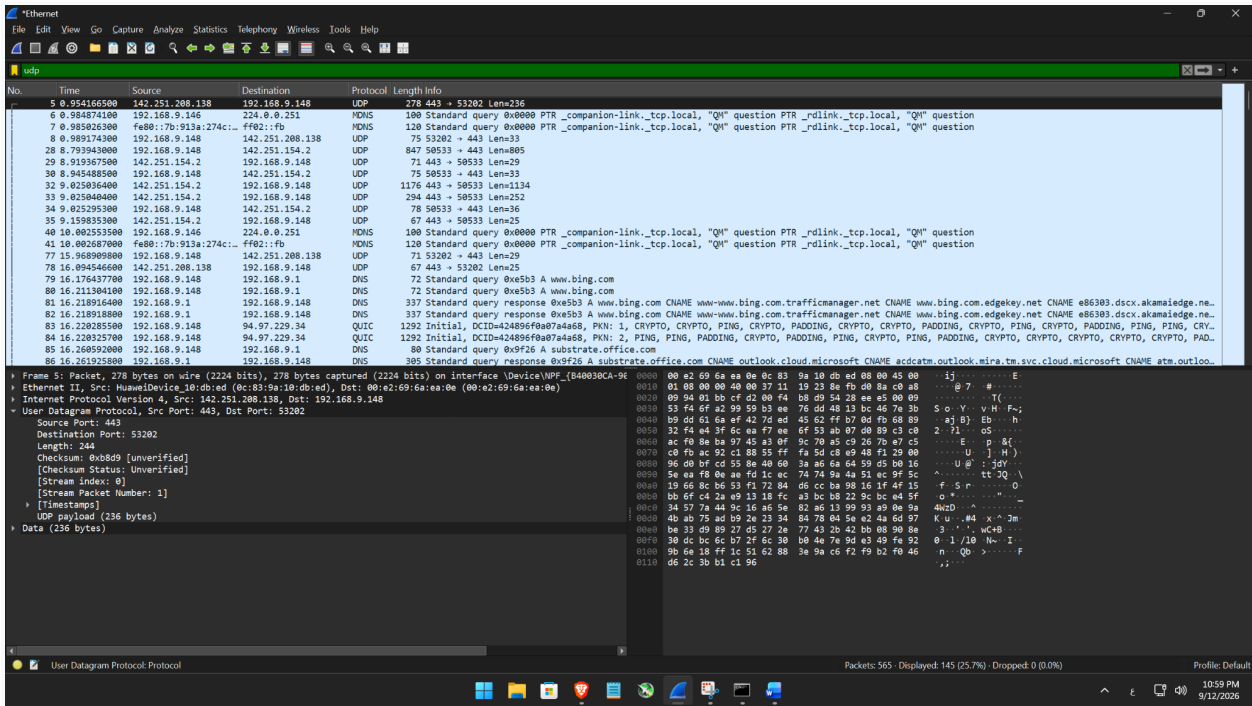
The bottom of the interface shows the packet summary: 'Packet 500.3 client pkts, 4 server pkts, 5 turns. Click to select.' The 'Find' bar is empty, and the 'Stream' tab is selected.

Part 3: Capturing and Analyzing UDP Traffic

UDP packets were captured and filtered using the display filter `udp`. Inspecting the User Datagram Protocol header demonstrated its lightweight, connectionless design compared to TCP:

- **Source Port:** 443
- **Destination Port:** 53202
- **Length:** 244 bytes
- **Checksum:** 0xb8d9

Unlike TCP, the UDP header consists of only 8 bytes across 4 basic fields, lacking sequence numbers, acknowledgment tracking, window size controls, or connection setup flags.



Part 4: Protocol Comparison Tables

Task 1: TCP vs UDP Characteristics

Criterion	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP is connection-oriented; it establishes a dedicated session via a Three-Way Handshake (SYN, SYN-ACK, ACK) and provides guaranteed delivery through acknowledgments, checksum validation, and automatic retransmissions. In contrast, UDP is connectionless and sends datagrams without establishing a session or guaranteeing delivery.
Data Integrity and Ordering	TCP	TCP guarantees ordered packet delivery and data integrity by utilizing Sequence Numbers to reassemble packets in the exact sequence they were sent, and discarding duplicates. While UDP verifies basic integrity using a checksum, it does not track packet sequence, handle reordering, or recover dropped packets.

Task 2: Use Cases and Performance

Aspect	TCP	UDP
Use cases	Web browsing (HTTP/HTTPS), secure shell (SSH), file transfers (FTP), and email communication (SMTP/IMAP) where data loss cannot be tolerated.	Real-time multimedia streaming, online multiplayer gaming, Voice over IP (VoIP), and DNS lookup queries where low latency is prioritized over complete reliability.
Performance	Lower throughput and higher latency due to connection setup overhead, substantial header size (20–60 bytes), acknowledgments, and congestion control mechanisms.	High throughput and ultra-low latency due to a minimal 8-byte fixed header, absence of connection establishment, and lack of retransmission delays.